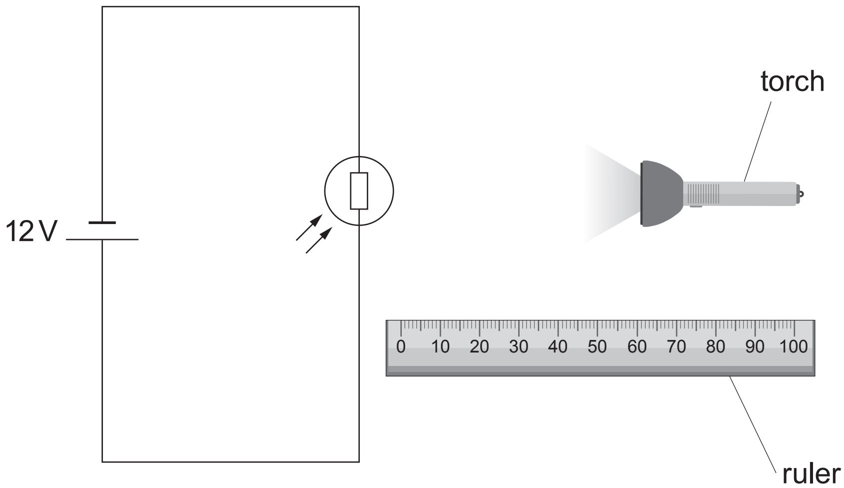
4. Dafydd, a year 10 student, set up a circuit to investigate the properties of a light dependent resistor (LDR). He changed the distance between a torch and the LDR. He then took measurements of current and voltage to calculate the resistance of the LDR.



(a) (i) **Complete** the circuit above by adding an ammeter to measure the current through the LDR and a voltmeter to measure the voltage across the LDR. [2]

(ii) The results of Dafydd's experiment are shown in the table below.

Distance of the torch from the LDR (cm)	Resistance of the LDR (Ω)
2	400
4	800
6	1000
8	1200
12	1300
16	1800
20	2000

Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the **current** passing through the LDR when the light source is 4 cm away. [2]

Current = ... A

(iii) Dafydd claims that for every 2 cm he moves the torch away from the LDR its resistance doubles. Use Dafydd's results to explain whether his claim is correct. [2]

....

....

....

....

(b) Emyr, another student, also carried out the experiment, but he used a 6 V battery. **Complete** the following sentence by underlining the correct phrase in the brackets. [1]

The resistances of the LDR in the table **(would be greater / stayed the same / would be less)**.

7

